

[MASK] [MASK] [MASK] [MASK]

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Diffusion [MASK] [MASK]

[MASK] [MASK] [MASK] Malyala, [MASK] [MASK]



CPSC 477 [MASK] [MASK]

[MASK] [MASK] Masking [MASK] Diffusion [MASK] [MASK]

[MASK] [MASK] Aniketh Malyala, [MASK] [MASK]



CPSC 477 [MASK] Project

[MASK] Guided Masking [MASK] Diffusion [MASK] [MASK]

[MASK] Feng, Aniketh Malyala, [MASK] [MASK]



CPSC 477 Final Project

Priority-Guided Masking [MASK] Diffusion [MASK] [MASK]

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CPSC 477 Final Project

Priority-Guided Masking For Diffusion Language Models

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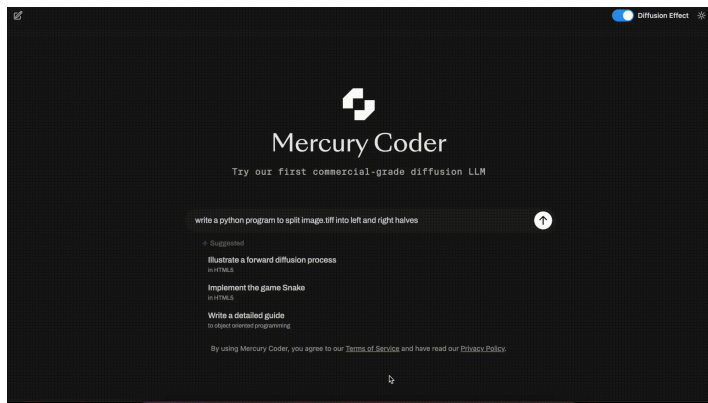
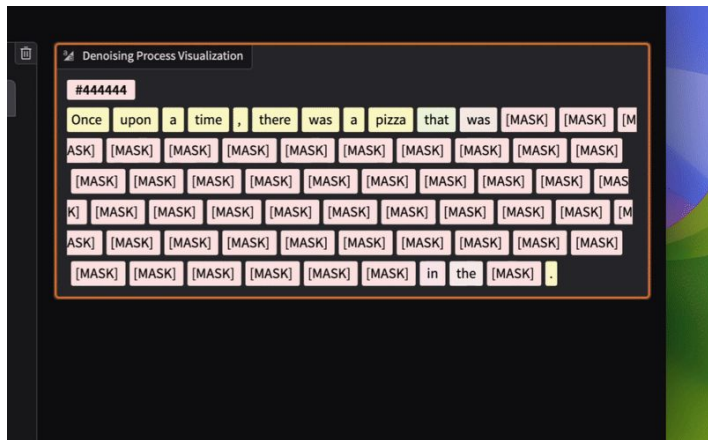


Introduction to Diffusion Language Models

- Diffusion models learn to generate data by reversing the process of adding noise
- DLMs adapt this idea from images to language
 - Forward process: turn text into masked tokens
 - Backward process: turn masked tokens into texts
- DLMs can refine entire sequences in parallel rather than autoregressively

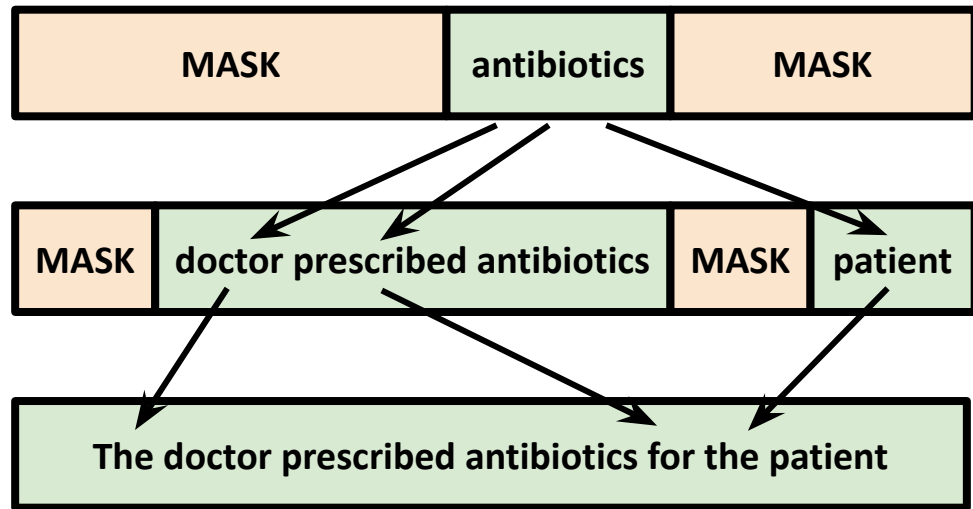
Leading DLMs: LLaDA, Mercury

- This approach has led to competitive DLMs
- [LLaDA](#): an 8B MDLM comparable to LLaMA 3 8B
- Mercury Coder by Inception Labs
 - Commercial MDLM
 - Up to [10x faster](#) than frontier speed-optimized LLMs



Motivation & LLaDA Training Scheme

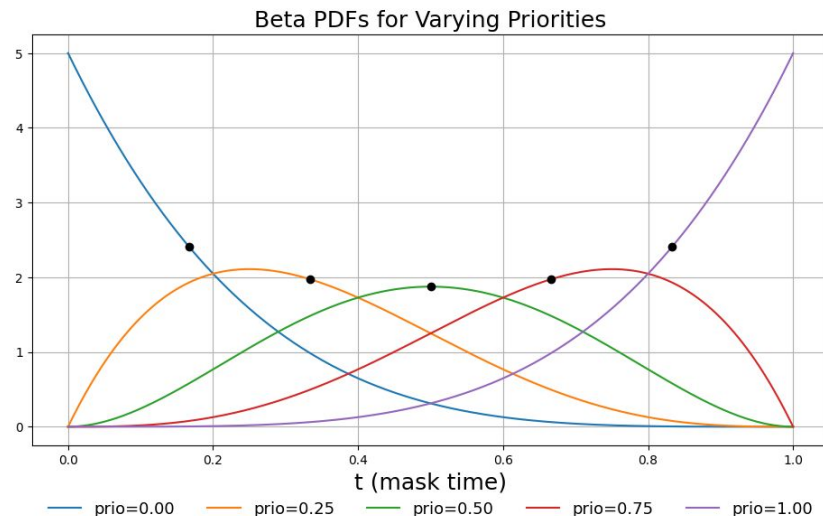
- Autoregressive models have a fixed order of generation
- DLMs can choose what order to generate text
- Certain words have more predictive power than others
 - antibiotics vs the
- Therefore we want to have “priority”-based generation



Claim: Generating keywords first will increase model performance

Modifying the Forward Process: Priority Masking

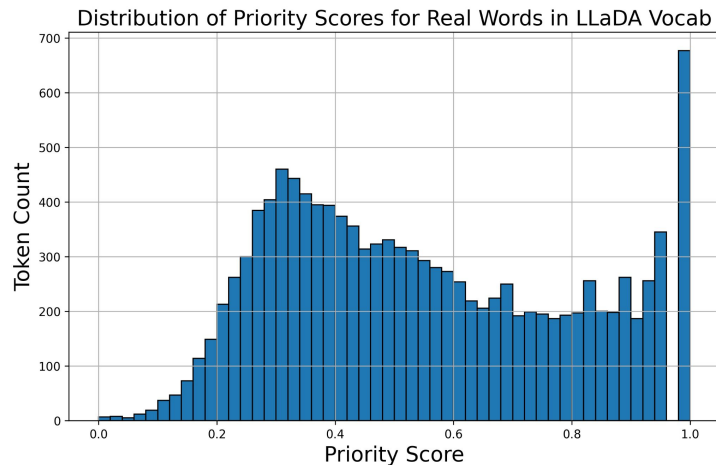
- LLaDA is trained by randomly masking some number of tokens and predicting the masked tokens
 - Each token is assigned $x \sim \text{Unif}(0,1)$ and tokens with $x < p \sim \text{Unif}(0,1)$ are masked
- We modify the process so that $x \sim \mu(x)$ so $\mu(x)$ is skew left (masked less often) when x has more priority
- $\mu(x)$ is $\text{Beta}(1 + \gamma p(x), 1 + \gamma(1 - p(x)))$ where $p(x)$ is x 's priority
 - Mean scales linearly with priority score
- We select masking probabilities, p , more biased towards 0.8-1



Assigning Priority Scores

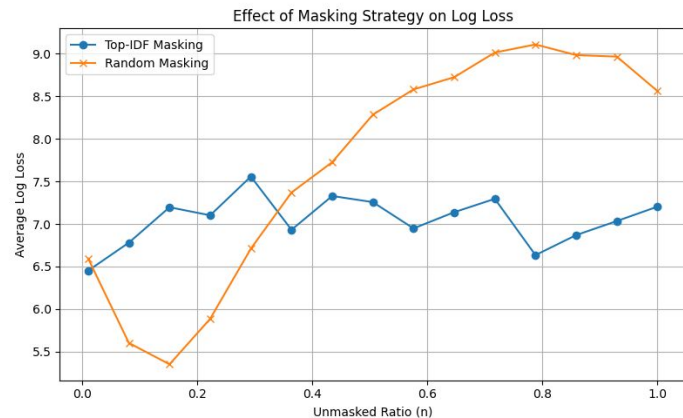
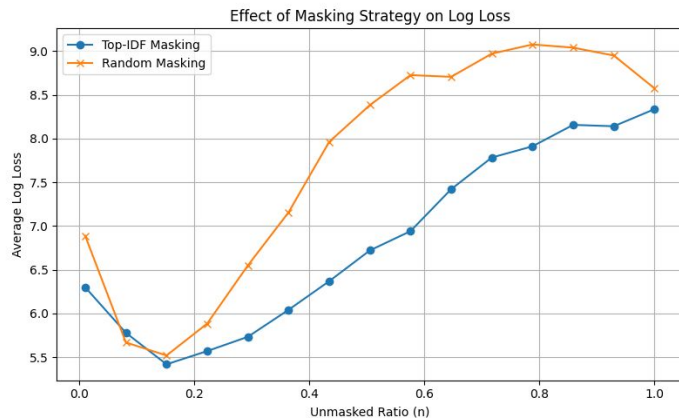
- Use Wikipedia IDF dataset that uses 6M article corpus
- 98.84% of LLaDA English vocab tokens (55K total) in dataset
- Min-max normalize priorities to be between 0 to 1
- Resulting scores are relatively balanced
- score < 0.05: the, is, a, to, by, was
- 0.5 < score < 0.7: slider, biden, reload, keywords
- score > 0.9: aleigh, omatic, -saharan

min	0.000000
25%	0.343003
50%	0.509629
75%	0.754607
max	1.000000



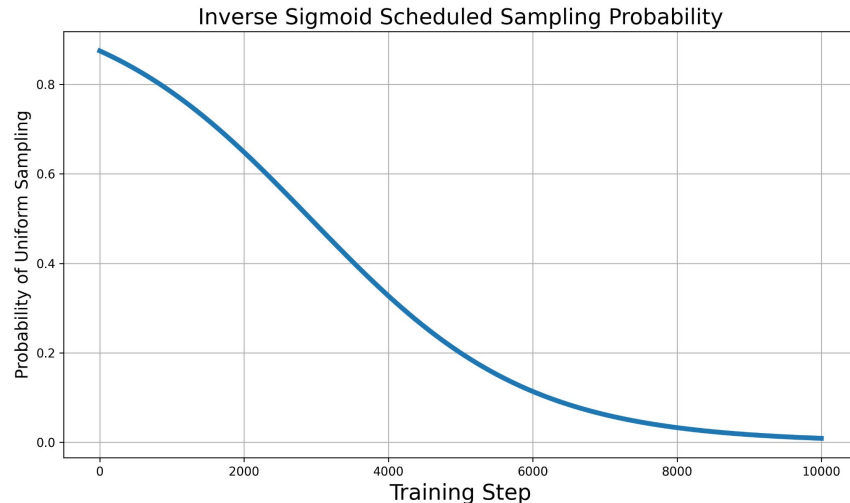
Exploratory Results

- We test our claim that generating keywords first boosts performance
- We mask tokens using 3 strategies and track loss on pretrained LLaDA
 - Random masking, mask top idf, mask top idf and resample 50%
 - We test on a varying number of masked tokens
- Top IDF + Resample is nearly always better than random masking
- Top IDF is significantly better when more tokens are unmasked



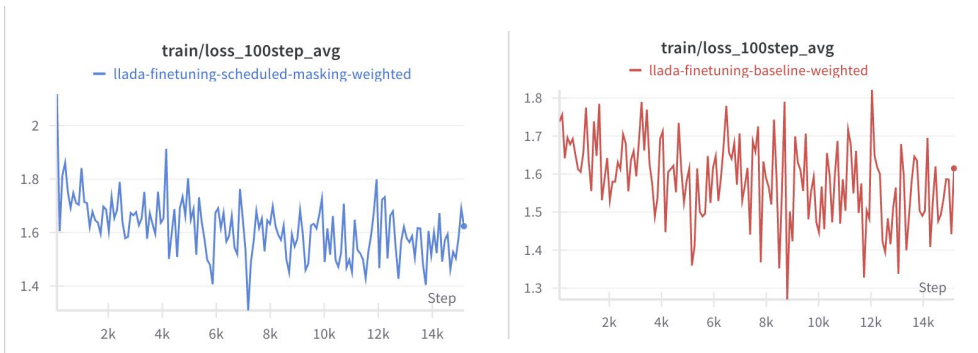
Fine-tuning Specifics

- Smoltalk instruction dataset (60k samples)
- Inverse Sigmoid scheduled sampling to transition from uniform masking to priority masking
- Finetune with purely uniform masking for baseline
- PeFT with LoRA
 - $r=8$, $\alpha=16$
 - $r=16$, $\alpha=16$
 - $r=32$, $\alpha=32$
 - **$r=64$, $\alpha=64$**



Fine-tuning Results & Evaluation

Training Loss & Validation Curves



Model	Validation Loss	Validation Perplexity
Scheduled Masking	4.854	128.313
Baseline	5.222	185.553

Testing Evaluations

Remasking Strategy Comparisons

Low Confidence	BLEU	ROUGE-1	ROUGE-L	BERTScore-F1
Pretrained Baseline	0.0138	0.0957	0.0833	—
Finetuned Baseline	0.1543	0.4989	0.3873	0.8923
Ours	0.1673	0.5060	0.3910	0.8931

Random	BLEU	ROUGE-1	ROUGE-L	BERTScore-F1
Pretrained Baseline	0.0109	0.0962	0.0824	—
Finetuned Baseline	0.1237	0.4785	0.3721	0.8898
Ours	0.1473	0.4954	0.3887	0.8932

- IDF averages **decrease more** in our model **as steps increase** than in the baseline model

Conclusion

Current Results

- Priority-guided masking is a promising path to improve performance
 - Log-loss experiments verify benefit of generating keywords early
 - Improvements over baseline when fine-tuning with our approach

Next Steps

- Part-of-speech based scores, learned scores, contextual scores
- Test different temperature settings, remasking strategies, other hyperparameters (LoRA settings, learning rates, etc.)
- Domain-specific fine tuning where keywords may be more important
 - Medical domain, law, etc.